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AI-driven skill keyword suggestion for multi-round interviews: A graph-based topic approach

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ABSTRACT

The application of artificial intelligence (AI) in recruitment has rapidly advanced in areas such as resume evaluation, person-job matching, and recommendation. However, AI's role in assisting interview questioning for skill assessment remains limited. This study proposes the Topic-Oriented Suggestion Model (TOSM), a deep learning framework designed to suggest skill assessment keywords tailored to multi-round interviews. The framework comprises two modules: a Graph-based Topic Learning Module, which captures latent relationships between skills and their contextual dependencies through graph representation learning, and a Topic-specific Keyword Suggestion Module, which dynamically adapts attention to the thematic topic focus of each interview round. Using a real-world dataset of multi-round interview cases from a leading technology company, TOSM achieves superior performance compared with baseline methods in keyword suggestion tasks. Visualization analyses further demonstrate that the model components effectively bridge vocabulary mismatches across different recruitment texts and adjusts its focus according to each interview round. This study contributes by proposing a novel framework that integrates graph-based topic model with multi-label learning. It extends AI-assisted recruitment research to the domain of interviewer questioning. It also provides practical implications for developing intelligent interview systems that support interviewer-side question design and improve the structure and efficiency of skill assessment in multi-round interviews.

1. Introduction

Artificial intelligence (AI) has ushered in the era of Digital Recruitment 3.0 (Black & van Esch, 2020), offering transformative opportunities for human resource management (HRM). Existing AI-driven recruitment research has largely focused on pre-interview decision support (e.g., resume screening, person-job matching, and job recommendation) (Albert, 2019; Pessach et al., 2020; Qin et al., 2018) or candidate-side interview assistance (e.g., automated scoring, individualized feedback, and coaching based on observable behaviors) (Chou et al., 2022; Lee & Kim, 2021). In contrast, our work targets interviewer-side support during multi-round interviews.

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The employment interview is the most critical phase of candidate evaluation for the company, and it typically consists of multiple rounds conducted by different interviewers. Interviewers play a vital role in the interview process because they take the responsibility for assessing the skills of the candidates effectively within a limited time window through their questioning strategies and assessment decisions, which influence interview reliability and validity (Judge et al., 2000). Recent studies have begun to explore interviewer training via post-interview intelligent systems (Yadav et al., 2023). However, AI support during the interview itself remains limited, particularly for targeted question design and application-context-specific skill assessment.

The human interviewer-based interview process presents two major challenges. The first is interviewer subjectivity. Variations in interviewers' experience, expertise, and questioning styles, particularly in unstructured interviews, can lead to inconsistent evaluation standards (Campion et al., 1997; Macan, 2009). When evaluation criteria vary across interviewers, interview ratings become less comparable, aligning with prior findings that unstructured interviews tend to exhibit lower reliability and validity than structured interviews with standardized questions and scoring criteria (Judge et al., 2000; McDaniel et al., 1994). The second challenge concerns redundancy across interview rounds. Although multi-stage interviews are designed to provide a more comprehensive evaluation, candidates often report that similar or overlapping questions are repeated in different rounds. This redundancy diminishes both the efficiency and the diagnostic value of the interview process (Huffcutt et al., 2001; Levashina et al., 2014). AI has demonstrated substantial potential to mitigate these issues by improving the consistency and transparency of human evaluation (Ganesan, 2020; Kleinberg et al., 2018). When applied as a supportive rather than a substitutive tool (Duan et al., 2019; Jia et al., 2024), AI-based approaches can help design more standardized but contextually relevant interview questions, thereby reducing subjectivity and redundancy across interview rounds. Motivated by this potential, this study aims to propose an AI-based framework that assists interviewers in questioning.

AI-based question generation and retrieval has achieved notable success in community question answering and educational testing (Figueroa, 2017; Zhang et al., 2023). These methods typically follow two steps: (1) given a keyword query, (2) generate or retrieve relevant questions based on that query. Because the query largely determines question quality, our framework focuses on the first step: providing precise, context-aware keyword queries to improve downstream AI-assisted interview questioning. Effective interview questions should align with both job requirements in postings and candidates' qualifications in resumes. We therefore formulate the task as providing skill assessment keywords from application information (job postings and resumes). Technically, applying this task to interviews involves two key challenges. First, skill expressions in application materials often differ from those used in interviews. As shown in Fig. 1, postings and resumes for an algorithm engineer may mention broad areas such as "deep learning", while interviews probe specific concepts (e.g., "random forest" or "word2vec") related to "deep learning". As a result, queries based on surface-level keywords may yield questions that are relevant in topic but shallow in depth. Second, multi-round interviews are a layered structure: each round targets different evaluative dimensions. In the same example, the first round often emphasizes prior experience (e.g., "kaggle game experience"); the second focuses on technical proficiency (e.g., "CRF" or "LSTM"); and the third evaluates personal fit and soft attributes (e.g., "engineering ability" and "smart"). In short, the three rounds prioritize experience, hard skills, and soft qualities, respectively. Each round thus requires tailored keyword queries to support round-specific questioning and candidate evaluation.

To address these challenges, we propose the Topic-Oriented Suggestion Model (TOSM), a novel AI framework designed to personalize the suggestion of skill assessment keywords for multi-round interviews. TOSM aims to standardize interview question generation by learning patterns from exemplary interview cases and aligning skill assessment criteria with interview stages. By leveraging graph-based topic modeling and attention-based multi-label learning, TOSM addresses vocabulary mismatches, adapts to layered questioning needs, and ensures comprehensive candidate evaluation. Utilizing these refined "key queries" suggested by TOSM, AI questioning tools like ChatGPT or retrieval systems can accurately access relevant questions from the question library. TOSM operates through two distinct stages: the Graph-based Topic Learning Module and the Topic-specific Keyword Suggestion Module. In the first stage, we employ a topic-oriented approach to identify the underlying connections between application materials and interview questions. This is achieved by using a graph-based technique to map the co-occurrence relationships among skill-related terms. Integrated with a neural topic model, this module elucidates potential semantic links between the textual representations of application skills and the graph representations of interview skills, effectively bridging their substantial expression gap of vocabulary mismatches. In the second stage, we enhance attention-based multi-label learning by transforming the original label-specific identification into topic-specific identification. We introduce the "Topic Query Attention" mechanism, enabling cross-text topic queries to capture key information in the application context more effectively than single-interview label queries. This module facilitates a multi-round focus, ensuring a comprehensive assessment across different interview stages.

In this work, we collect large-scale metadata about job postings, resumes, and interview questioning records from an IT company's recruitment department. Based on these raw text documents, we construct a real-world dataset and conduct extensive experiments to evaluate our proposed approach. Initially, we extracted skill-related terms from the dataset to enhance our model's focus on skill representation learning. TOSM was then trained using data from exemplary interview cases, enabling it to align skill assessment criteria with the typical interview pattern, focusing sequentially on experience, hard skills, and soft skills across three rounds. To verify the superiority of TOSM and the effectiveness of each module in predicting skill assessment keywords, we selected several competitive methods of representation learning and prediction as well as designed the variants of TOSM for comparison. Experiments have proved that in the metrics of *precision* and *NDCG*, our TOSM outperforms these comparison methods. Additionally, we conduct a visual analysis of TOSM to illustrate the impacts achieved by each module within the framework. Lastly, we design a question generation case utilizing ChatGPT to juxtapose questions generated using TOSM-suggested skill words against those based on the original application information.

This study makes both theoretical and practical contributions. Methodologically, it develops a novel TOSM that integrates graph-based topic learning with topic-oriented multi-label learning to generate context-aware skill assessment keywords for multi-round

JOB POSTING

-Strong problem analysis and solving skills in computer vision or natural language understanding
-Familiar with at least one deep learning framework and the most cutting-edge models, algorithms in image processing and NLP
-Have experience in code programming
-Understanding and decision-making capabilities in the face of massive user and image search requests

RESUME

-Articles recommendation based on word2vec, model is parsed by Scala and finally deployed as a web service
-Kaggle data science competition——plankton image classification based on deep learning, classify images through convolutional neural networks, and extract image feature through various pre-trained feedforward neural networks
-New word discovery based on CRF, build sequence labelling models, the model file is parsed through Scala, and the distributed version uses Spark to optimize the model

Skills in Job Posting	Skills in Resume
problem analysis and solving	
computer vision image processing	image classification extract image feature
natural language understanding NLP	Articles recommendation word discovery
deep learning cutting-edge models algorithms	Word2vec deep learning Convolutional neural networks feedforward neural networks CRF
code programming	
	Kaggle data science competition
Understanding and decision-making capabilities	web service Scala distributed Spark

Key Skill Assessment Points of Interview Record	
Round1	kaggle game experience similar articles recommended NLP-related concepts programming questions
Round2	data structure algorithms reverse array scripting language coding word2vec, random forest, GBDT principle, evaluation method, common loss basic concepts of HMM, CRF, LSTM machine learning fundamentals learning ability
Round3	engineering ability problems solving smart

Skill words with the same thematic topic are highlighted in the same color.

Fig. 1. An example of an algorithm engineer's skill keywords from job posting, resume, and interview records.

interviews. The framework effectively bridges semantic gaps among job postings, resumes, and interview records while adapting to the evolving focus of different interview rounds. Theoretically, this research extends AI-driven recruitment and interview by shifting attention from pre-interview recommender systems and candidate-side interview AI to interviewer-side questioning support during the interview. In doing so, it offers a design perspective for studying how AI suggestions can be integrated into interviewer questioning and multi-round skill assessment. Practically, it offers design insights for developing AI-assisted interview systems, such as question recommendation and interviewer-candidate matching tools, that promote more structured and efficient interview assessment.

The remainder of this paper is organized as follows. Section 2 provides a detailed review of the existing literature, covering prior studies on AI-based interview as well as related methodological approaches. In Section 3, we define our problem and introduce the proposed TOSM and elaborate on its two key components: the Graph-based Topic Learning Module and the Topic-specific Keyword Suggestion Module. Section 4 details the dataset, data processing techniques, and experimental setup. We evaluate the performance of TOSM through extensive comparisons with baseline methods, conduct an in-depth analysis of its key components, and perform robustness checks. Section 5 discusses the contributions of our study, practical applications, limitations, and potential directions for future research.

2. Literature review

In this section, we first review the literature on interview in AI-driven recruitment and background of keyword-based questioning, then review the methodological foundations of our approach, including topic model and multi-label learning.

2.1. Development of AI-based interview

In the knowledge-driven economy, intangible assets such as knowledge, skills, and innovation have become key drivers of enterprise growth (Brinkley, 2006). As the importance of specialized knowledge increases, recruiting employees with job-fit skills has become essential for maintaining competitiveness (Hamilton & Davison, 2018). Accordingly, data-driven HRM research has leveraged large-scale job and resume datasets to support talent selection in the pre-interview stage, including person-job matching, resume screening, job recommendation, skill extraction, and labor market analysis (Chunmian et al., 2021; Kokkodis & Ipeirotis, 2021; Meng et al., 2022; Zhu et al., 2018).

Despite these advances, interviews remain a long-standing and central practice in hiring, commonly defined as a "face-to-face interaction conducted to assess an individual's suitability for a given job". The HRM literature has long examined interview effectiveness, focusing on interview format (e.g., individual vs. panel), interview structure (e.g., unstructured vs. structured), and interviewer individual differences as key sources of variation in interview reliability and validity (Hartwell et al., 2019; Judge et al., 2000; McDaniel et al., 1994). More recently, digital technologies have reshaped how interviews are conducted. Web-based interviews reduce logistical costs and support both synchronous and asynchronous formats to aid analysis and fair selection (Brenner et al., 2016). Building on these digital formats, an emerging stream of research has introduced AI-enabled tools into interviews. Most existing work focuses on the candidate side, which analyzes candidates' behavioral signals (e.g., facial expressions, vocal tone, and body language) to predict ratings or provide feedback to improve interview performance (Chou et al., 2022; Lee & Kim, 2021). In contrast, AI-enabled support for interviewers remains comparatively limited. Although some studies have explored interviewer-side tools, such as training systems and post-interview feedback (Yadav et al., 2023), real-time support during the interview is still underdeveloped, especially for targeted question design and application-specific skill assessment.

2.2. Background of keyword-based questioning

In the natural language processing domain, extensive research has been conducted on automatic questioning, particularly in question generation and question retrieval. These methods are widely used in question-answering systems, search engines, and recommender systems for community QA, open source code platforms and education exercises (Abdelghani et al., 2023; Figueroa, 2017; Sychev et al., 2023). Among these techniques, keyword-based methods play a central role, as the accuracy of keywords directly affects the relevance of generated or retrieved questions. For example, Sajjad et al. (2012) extracted key terms from unstructured text for query refinement, while Kim et al. (2019) employed a knowledge graph-based keyword extraction model for question generation.

For non-expert users, textual queries are the most intuitive way to interact with retrieval or generation systems. However, users often struggle to articulate precise queries due to limited domain knowledge. Extensive research has explored this problem through query suggestion, expansion, and refinement (Ooi et al., 2015). The issue of vocabulary mismatch, a prevalent problem where semantically similar terms differ across documents on the same topic, has been explored using graph structures (Shekarpour et al., 2017) and common topic spaces (Anjum et al., 2019), particularly receiving attention in the medical domain (Goeuriot et al., 2016). A similar challenge arises in recruitment interviews, where differences in linguistic representation across job descriptions, resumes, and interview discussions can reduce the effectiveness of keyword queries in identifying questions relevant to candidate evaluation.

2.3. Topic model

Topic models excel at uncovering hidden semantic structures by mapping high-dimensional word spaces to low-dimensional topic spaces, facilitating a quick and accurate understanding of text information (Belwal et al., 2021; Kriebel & Stitz, 2022). Anjum et al. (2019) proposed a common topic model for solving the problem of vocabulary mismatch in paper-reviewer matching, which gave us insights. Latent Dirichlet Allocation (LDA) (Blei et al., 2003) and its extensions (Newman et al., 2010; Rosen-Zvi et al., 2012) specify generative models for documents using latent variables sampled from prior distributions and word distributions as topics. Recent Neural Topic Models (NTMs), such as GSM (Miao et al., 2017), NVDM (Miao et al., 2016), and AVITM (Srivastava & Sutton, 2017), operate within the Variational Autoencoder (VAE) framework (Kingma & Welling, 2013), offering effective inference. NTMs capture intricate semantic associations between latent topics and words, diverging from surface-level analysis by LDA-based methods (Chunmian et al., 2021; De Mauro et al., 2018; Li et al., 2023). To model the complicated dependencies between topics, Huang et al. (2024) built the DepNTM on a VAE framework with a DAG, which encodes the documents and their labels into latent topical dependency representations. Beyond classical probabilistic topic models (e.g., LDA) and NTMs, recent studies have explored transformer-based topic modeling, which derives topics by clustering document representations from pretrained language models (Grootendorst, 2022). Such approaches often produce highly coherent semantic themes for free text. However, for tasks that map heterogeneous texts to a shared label space, it is often necessary to model label-level dependencies (e.g., co-occurrence or hierarchical taxonomy) in addition to document semantics. Purely transformer-based topic clustering may not respect such label structure, motivating topic modeling approaches that integrate relational information. NTMs' flexibility allows deep semantic mining customization. For instance, a relational topic model-based approach for bilateral review analysis was developed by Chen et al. (2021), while Liu et al. (2019) enhanced word representation by incorporating latent topic information into word-level semantic representation for sentiment analysis. Building on these advances, this study adopts GSM (Miao et al., 2017) as the foundational model and enhances it with graph-based representation learning to improve cross-text semantic alignment in recruitment data.

2.4. Multi-label learning

Multi-label learning is widely applied across domains such as text categorization, image annotation, information retrieval, and recommendation systems (Agrawal et al., 2013). In addition to conventional methods like one-vs-all (Babbar & Schölkopf, 2017), tree-based classifiers (Prabhu & Varma, 2014), and embedding-based models (Bhatia et al., 2015; Tagami, 2017), deep learning approaches are gaining prominence in multi-label learning. Attention mechanisms play a key role in enhancing multi-label learning. Attention-based multi-label learning focuses on deriving label-specific document representations from content (Huang et al., 2021; Song et al., 2022; Xiao et al., 2019). Attention mechanisms establish semantic connections between label-specific representations and document content (Huang et al., 2022), capturing semantic relationships. Some methods expand to label dependencies by constructing graphs or hierarchical structures (Cai et al., 2020; Cheng & Shi, 2025; Huang et al., 2019; Tang et al., 2020), utilizing Graph Convolutional Networks (GCN). Some advanced methods involve label graph (Vu et al., 2023) into Bert-based learning (Adhikari, 2019).

However, existing label-word attention mechanisms are not directly applicable to our problem. In interview scenarios, labels represent skill-related terms from interview questions, while textual data come from job postings and resumes. The disparity in surface-level semantics makes it difficult to establish direct label-word correspondences. To address this challenge, we introduce a topic-guided attention mechanism, termed “Topic Query Attention”, which combines topic modeling and multi-label learning to capture latent semantic relations between interview and application expressions. This design improves topic-specific representation learning and supports more accurate skill keyword prediction.

2.5. Summary and research gaps

In summary, existing studies on AI-assisted recruitment have mainly focused on early-stage tasks such as resume screening, person-job matching, and recommendation, while paying limited attention to interviewer support during the interview questioning process. Our task is centered on the core of question retrieval and generation, namely keyword query. Although keyword-based questioning has been widely studied, the persistent issue of vocabulary mismatch also exists among job descriptions, resumes, and interview questioning. This study addresses this gap by improving keyword query accuracy for interview questioning. Furthermore, the fundamental methodologies used in prior work, topic modeling and multi-label learning, are typically designed for homogeneous texts with static label dependencies, making them unsuitable for cross-text and multi-round interview settings. Building on these gaps, this study advances topic modeling and multi-label learning to develop novel methods tailored for AI-assisted interviewing.

3. Our proposed framework TOSM

In this section, we propose a TOSM to predict suggested skill assessment keywords by learning the previously accumulated interview-related data from exemplary interview cases. We first define the problem and outline the framework, and then describe each component in detail.

3.1. Problem definition

Assume that our dataset contains $|M|$ position-applicant-interview unique triplet instances, i.e., $T = \left\{t_m = (j_m, r_m, i_m)\right\}_{m=1}^{|M|}$, where each element is abbreviated as $t = (j, r, i)$ and j, r, i respectively represents job posting, resume and interview questioning records. In order to reduce unnecessary input noise and focus on the perspective of professional ability and competence, we only consider the skill-related words in the raw data. We established skill-related vocabulary for each collection in the recruitment dataset, $\mathbf{V_J}$ for job posting, $\mathbf{V_R}$ for resume, and $\mathbf{V_I}$ for interview questioning records. For a job posting j, we denote $j = \left\{w_1^j, w_2^j, \ldots, w_{N_j}^j\right\}$, where $w_*^j \in \mathbf{V_J}$ is the $*$-th skill in j. Similarly, for a resume r, we denote $r = \left\{w_1^r, w_2^r, \ldots, w_{N_r}^r\right\}$, where $w_*^r \in \mathbf{V_R}$ is the $*$-th skill in r. In particular, one application may have multiple rounds of interview with D interview questioning records $i = \{i_1, \ldots, i_D\}$, and in our experiment $|D| = 3$. So for an application's d-th round interview record i_d, we denote $i_d = \left\{w_{d,1}^i, w_{d,2}^i, \ldots, w_{d,N_{i_d}}^i\right\}$, where $w_*^i \in \mathbf{V_I}$ is the $*$-th skill in i_d. To suggest the skills evaluated in interviews, we formulate the assessment point suggestion as a multi-label learning task. Our goal is to predict the skill word set i_d for each interview round, where each skill word in i_d is treated as a label.

3.2. Framework overview

As illustrated in Fig. 2, the overall framework TOSM has two key components: a Graph-based Topic Learning Module and a Topic-specific Keyword Suggestion Module. In the Graph-based Topic Learning Module, we treat application information (job posting and resume) as prior knowledge and interview questioning records as posterior knowledge. We represent the prior using a bag-of-words (BOW) vector and the posterior using a skill graph, which better captures dependency relations among skill terms in interview records. A Bayesian latent variable model is established between the prior and the posterior knowledge, mining the underlying semantic topic connections between application information and the interviewer's questioning record. Emphasize that integrating the graph into the NTM as posterior knowledge is the most significant difference between our approach and the traditional topic models or NTMs. Based on the learned topic connections between prior and posterior knowledge, in the Topic-specific Keyword Suggestion Module, we develop a multi-label learning task with a topic-oriented attention mechanism to predict skill assessment keywords depending on the input application information. We first describe the skill graph construction and then present the technical details of the two modules.

3.3. Skill graph representation of interview questioning record

We construct a skill graph based on the co-occurrence relationships among skill words in interview questioning records. Given the sequential nature of skill words extracted from the interview texts, we define edges between words that appear together within a fixed window. For instance, although “word2vec” and “GBDT principle” are not semantically similar, they often appear consecutively in interviews, indicating a non-semantic association.

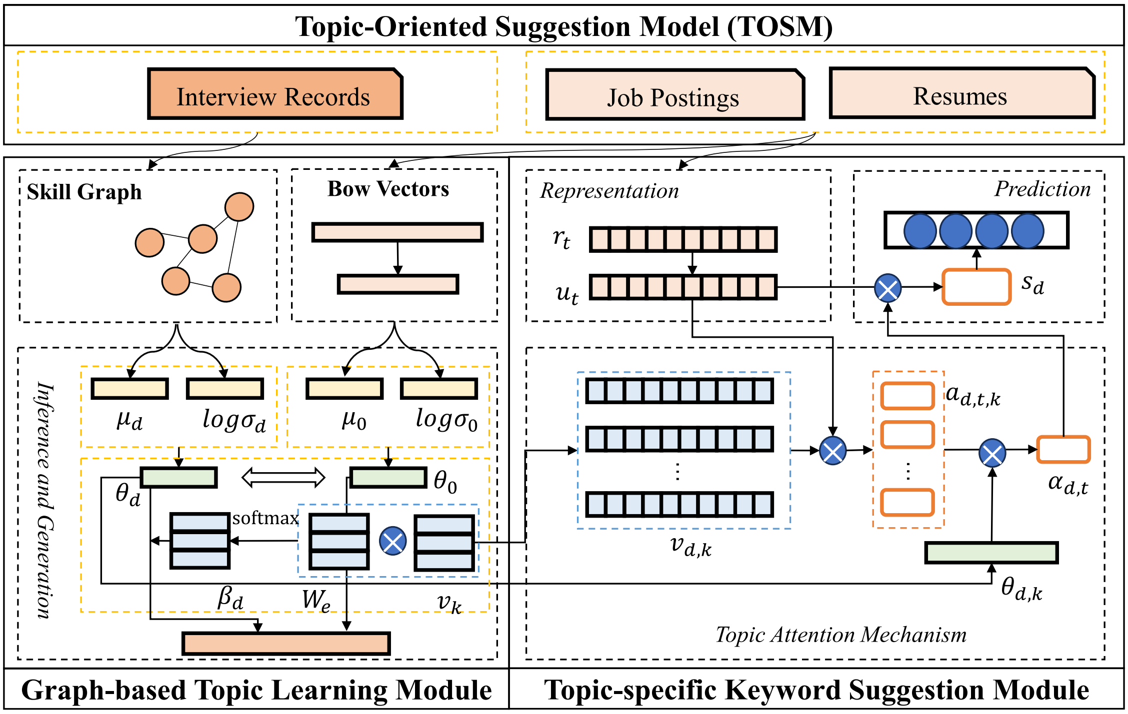


Fig. 2. Overview of TOSM structure.

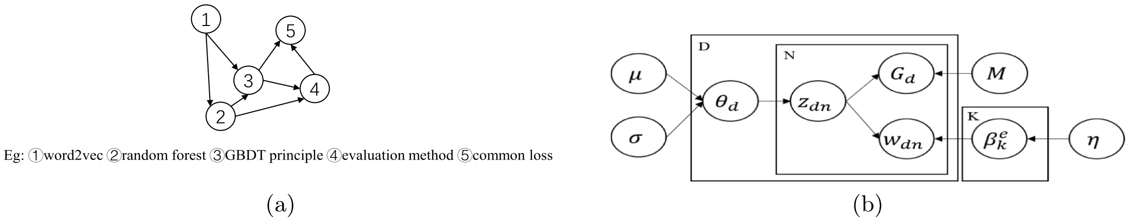


Fig. 3. (a) The graph construction example for sequence of skill words in an interview questioning record when $l = 2$, (b) the probabilistic graph of the Graph-based Topic Learning module.

Let the sequence of skill words in an interview questioning record be denoted as $i_d = [w^i_{d,1}, \dots, w^i_{d,N_{i_d}}]$, where each $w^i_{d,s}$ represents a skill word. We construct a directed edge from w^i_{d,s_1} to w^i_{d,s_2} if $0 < s_2 - s_1 \le l$, where l is the window size. Thus, each interview questioning record's skill words are represented as a graph $G_d = (i_d, E_d)$, where i_d is the set of words, and E_d is the set of edges between words.

Fig. 3(a) shows an example of constructing a skill graph with $l = 2$. Nodes represent skill words such as ① "word2vec", ② "random forest", and ③ "GBDT principle". Edges capture the co-occurrence of words within the window size. For instance, the edge between ① and ② indicates that "word2vec" and "random forest" appeared together within the window. This graph structure preserves the sequential nature of interview questions and contextual dependencies. To reduce noise, we remove low-frequency skill words and dependency edges. Unlike traditional topic modeling methods that treat text as an unordered set of words (e.g., BOW), our graph-based approach captures word relationships that reflect their sequential order. This is also useful for multi-label learning: modeling skills as a graph helps capture complex, non-semantic dependencies among labels. Compared with prior label-graph embedding or attention methods, our skill graph better reflects such non-semantic label relations observed in interviews.

3.4. Graph-based topic learning module

Here, we design a generative topic model for mining the semantic connections between the application information and the interview questioning record. Similar to other topic models (Blei et al., 2003), we assume there exist K topics, i.e, $\beta = \{\beta_k\}_{k=1}^K$, in the interview record collection. Each topic k is a multinomial distribution β_k on vocabulary V_I. As shown in Fig. 3(b), for each interview record i_d, we define the joint possibility of the document graph G_d, topic proportion θ_d and topic assignment set $Z_d = \{z_{d,n}\}_n$ as follows with the job posting and resume as the conditional information,

$$p(G_d, \theta_d, Z_d | j, r) = p(\theta_d | j, r) p(E_d | Z_d) \prod_{n=1}^{N_{i_d}} p(z_{d,n} | \theta_d) p(w^i_{d,n} | z_{d,n}, \beta). \tag{1}$$

In the following, we will first specify each factor in the above equation with the detailed generative process and then introduce the model inference process with detailed neural network structures.

3.4.1. Generation process

Here, we aim to formulate the process of generating the interview record i_d. As shown in Fig. 3(b) and Eq. 1, we design the generative process as follows:

$$\theta_d \sim p(\theta_d | j, r) = \mathrm{GN}\big(\mu_0, \sigma_0^2\big), \text{ for } d \in D, \tag{2}$$

$$z_{d,n} \sim p(z_{d,n} | \theta_d) = \text{Multi}\left(\theta_d\right), \text{ for } n \in \left[1, N_{i_d}\right], \tag{3}$$

$$w^i_{d,n} \sim p(w^i_{d,n} | z_{d,n}, \beta) = \text{Multi}\left(\beta_{z_{d,n}}\right), \tag{4}$$

$$E_d \sim p\big(E_d \mid Z_d\big), \tag{5}$$

where $\mathrm{GN}\big(\mu_0, \sigma_0^2\big)$ is composed of a neural network $\theta = g(x)$ conditioned on a Gaussian random variable $x \sim \mathcal{N}\big(\mu_0, \sigma_0^2\big)$ and the activation function in the last layer in $g(\cdot)$ is the softmax function. In particular, noting the strong connections between application information, i.e., the job posting and resume, and the interview record, we combine j and r together as the tuple (j, r) represented by a for an application profile and regard a as the prior knowledge. We design this generative process with the motivation that the job posting and resume provide strong prior cues for the underlying skill topics in the interview records. By incorporating the application information into the generation of θ_d of the document d, we ensure that the inferred topics are aligned with the applicant's personalized information. Specifically, we assume μ_0 and σ_0^2 are determined by (j, r) with a neural network, i.e.,

$$\mu_0 = g_1(\hat{a}), \quad \log(\sigma_0^2) = g_2(\hat{a}), \tag{6}$$

where $g_1(\cdot)$, $g_2(\cdot)$ are Multilayer Perceptron networks with batch normalization and drop-out, and $\hat{a}$ is the concatenation of the BOW vectors of j and r.

After sampling the topic proportion θ_d, we assume that each word $w_{d,n}$ (or placeholder n) has corresponding topic assignment $z_{d,n}$ drawn from $\text{Multi}(\theta_d)$. Then, the word $w_{n,d}$ can be sampled from the corresponding word distribution $\beta_{z_{d,n}}$ of topic $z_{d,n}$. However, different from the previous topic models, we also involve the generation of graph structure to capture the word dependency in documents. Specifically, we assume that the edge set E_d is drawn only conditionally on the topic assignment set $Z_d = \{z_{d,n}\}_n$, with the assumption that the word dependency is based on the semantic dependency among topics. In sum, we formulate the $p(E_d \mid Z_d)$ as the joint possibility of both existing edges and non-existing edges among words, i.e.,

$$p(E_d \mid Z_d) = \prod_{(n,n') \in E_d} m_{z_{d,n}, z_{d,n'}} \prod_{(n,n') \neq E_d} (1 - m_{z_{d,n}, z_{d,n'}}), \tag{7}$$

where $m_{k,k'} \in [0, 1]$ represents the possibility of the connection of two words with topic k and k'.

3.4.2. Inference process

Inspired by NTMs, we infer our model by applying neural variational inference. Formally, we define the variational family with the interview record as posterior knowledge as conditional information.

$$q(\theta_d, Z_d | G_d) = q(\theta_d | G_d) \prod_{n=1}^{N_{i_d}} q(z_{d,n} | G_d, w^i_{d,n}), \tag{8}$$

where $q(z_{d,n} | G_d, w^i_{d,n})$ is a multinomial distribution with parameter $\varphi_{d,n}$, and $q(\theta_d | G_d)$ is specified as the Gaussian-based distribution $\mathrm{GN}(\mu_d, \sigma_d^2)$ for approximating to the true posterior of θ_d with the prior $\mathrm{GN}(\mu_0, \sigma_0^2)$.

To be specific, here, we need to model the mapping from the observed interview skill data G_d to the variational parameters $\mu_d, \sigma_d, \varphi_{d,n}$ for latent variables θ_d and z_d. We denote $X_d \in \mathbb{R}^{|V_I| \times L}$ as the trained and fixed word embeddings for all words in the whole vocabulary V_I. Then, we encode the local information of each node and the global information of the whole graph G_d with one layer of graph neural network, such as GraphCov (Welling & Kipf, 2016):

$$h_d = \tanh(GNN(G_d, X_d)), \quad h_d^G = \tanh\left(\sum_{n=1}^{N_{i_d}} \sigma\big(f_1\big(h_{d,n} \oplus x_{d,n}\big)\big) \odot \tanh\left(f_2\big(h_{d,n} \oplus x_{i,n}\big)\right)\right), \tag{9}$$

where $h_d \in \mathbb{R}^{|V_d| \times L}$ is the node-level representation vectors, $h_d^G \in \mathbb{R}^L$ is the graph-level representation vector, and $f_*(\cdot)$ is a full connected layer. Self-loops have been added to G_d to preserve the current word feature. In our design, encoding both local (node-level) and global (graph-level) information is crucial. In an interview scenario, the local node-level features capture how closely related skills appear together in short windows, reflecting the immediate context of the interviewer's follow-up questions. By contrast, the global representation h_d^G encodes the overall theme or skill distribution of the entire record. This is crucial because interviewers often pivot from one skill cluster to another depending on the candidate's background or the job requirements, thus introducing a more holistic structure to the questioning. Consequently, combining node-level (local) and graph-level (global) embeddings allows the model to mirror realistic interviewing patterns. This dual encoding enhances the model's ability to accurately predict the key skill assessments.

Then, the parameters of variational marginals $q(\theta_d \mid G_d)$ and $q\left(z_{d,n} \mid G_d, w^i_{d,n}\right)$ can be specified as:

$$\mu_d = f_\mu(h^G_d), \quad \log\sigma^2_d = f_\sigma(h^G_d), \quad \varphi_{d,n} = f_\varphi(h^G_d \oplus h_{d,n} \oplus x_{d,n}). \tag{10}$$

After defining the generative process in Eq. 1 and Eq. 8, we can optimize the Evident Lower Bound (ELBO) between them to infer the model parameter, i.e.,

$$\begin{aligned}\mathcal{L}_d =& E_{q(Z_d|G_d)}[\log p(E_d|Z_d) + \sum_{n=1}^{N_{i_d}} \log p(w^i_{d,n}|z_{d,n},\beta)] \\ & - KL[q(\theta_d|G_d)||p(\theta_d|j,r)] - E_{q(\theta_d|G_d)}[\sum_{n=1}^{N_{i_d}} KL[q(z_{d,n}|G_d,w^i_{d,n})||p(z_{d,n}|\theta_d)]],\end{aligned} \tag{11}$$

where the first line aims to optimize the reconstruction of the skill graph G_d for interview record i_d, and the second line is to minimize the Kullback-Leibler (KL) distance between the prior and posterior of latent variables, i.e., θ_d and $z_{d,n}$. Fortunately, all terms in the above equation are tractable. Then, we can optimize it to infer the best parameters of our topic model and extract the latent semantic connection between applicant information and interview records. This process will enhance the skill assessment suggestion model in the following section.

3.5. Topic-specific keyword suggestion module

3.5.1. Representation of application information

We use BiLSTM to model word representation in job postings and resumes. For an application profile, i.e, the tuple of job posting and resume $a = (j, r)$, we denote the word sequence in the concatenation of them as $a = \left\{s^a_1, s^a_2, \dots, s^a_{N_a}\right\}$, where the symbol s^a_i also denotes its one-hot vector representation. Then, we embed the words into vectors by $w_t = W_e s^a_t$, $w_t \in \mathbb{R}^{d_0}$, where w_t denotes d_0-dimensional word embedding of t-th word in a. The matrix W_e is initialized by a pre-trained word vector matrix and re-trained during the training process. Then for each word in the m-th application, we can calculate the representation $\left\{r_1, r_2, \dots, r_{N_a}\right\}$ by:

$$r_t = BiLSTM\left(w_{1:N_a}, r_{t-1}\right), \tag{12}$$

where $w_{1:N_a}$ denote the word vectors of input sequences of a.

3.5.2. Topic attention mechanism

The attention mechanism was originally proposed by (Bahdanau et al., 2014) in machine translation. The attention mechanism calculates the similarity between a context vector (query) and each key (word) to obtain the attention score corresponding to the key. The trainable context vector can be seen as a high-level representation of a fixed query "what is the informative word" in the sequence.

Recall that in neural topic models, each topic $\beta_k \in \mathbb{R}^{|V_I|}$ is a multinomial distribution over words. However, β_k is not regarded as a single parameter matrix in the generative process and can be written as the product of two smaller learnable matrices, i.e., the topic vector $v_k \in \mathbb{R}^{d_0}$ and the word matrix W_e:

$$\beta_k = \mathrm{softmax}(v_k W_e). \tag{13}$$

In our topic model, instead of using only one context query, we use multiple topic vectors as queries. These queries learned by the Graph-based Topic Learning Module can help the suggestion to focus on the information regarding to different aspects of the corpus from a global topic perspective, which may otherwise be overlooked by the LSTM model that mostly focuses on the local contextual information. An intuitive explanation is that these topic vectors are used as queries to resemble "what is the informative word under this topic". After calculating attention weights concerning all the topics, we then average them by a shifted topic proportion, which not only makes the final attention weights emphasize the most important topics but also diversifies different topical attentions by backpropagation.

For the topic vectors gained from the d-th interviewing round, $\{v_{d,k}\}^{K_d}_{k=1}$, and step $t \in \{1, 2, \cdots, N\}$, the i-th topical attention weight at the t-th step is:

$$u_t = \tanh(W_t r_t + b_t), \quad a_{d,t,k} = \mathrm{softmax}_k(u^T_t v_{d,k}), \tag{14}$$

where u_t has the same dimension as $v_{d,k}$. The topic vectors $v_{d,k}$ are learned parameters via the Graph-based Topic Learning Module, serving as a critical bridge between two modules, as shown in TOSM structure in Fig. 2. Consequently, the keyword suggestions are also aligned with the semantic themes present in the interview records. This unified approach addresses limitations in previous work

Table 1
Basic information of the source datasets.

Dataset	Data volume	Contained information
Position	5,913	job posting ID, application ID, job type, job title, job requirements, job content
Applicant	18,856	resume ID, application ID, education background, project experience
Interview Report	71,726	interview ID, application ID, interview type, interview result, evaluation record

that treated topic modeling and keyword suggestion as separate tasks, resulting in more coherent keyword predictions in interview scenario. To put the K_{d} topical attentions at the t-th step together as a scalar, we average them by a shifted topic mixture:

$$\alpha_{d,t} = \sum_{k=1}^{K_{\mathrm{d}}} a_{d,t,k}\theta_{d,k}. \tag{15}$$

By applying this set of attention weights to the hidden states, we get the final encoded document in d-th round:

$$s_d = \sum_{t=1}^{N_a} \alpha_{d,t} r_t. \tag{16}$$

3.5.3. Prediction and loss

We apply binary cross-entropy loss as the loss function in our suggestion model. The binary cross-entropy loss function in our suggestion model can be formulated as follows:

$$loss_d = -\frac{1}{|M|}\sum_{m=1}^{|M|}\sum_{k=1}^{|L|}\left[y_{m,d,k}\log\sigma\left(s_{m,d,k}\right) + \left(1 - y_{m,d,k}\right)\log\left(1 - \sigma\left(s_{m,d,k}\right)\right)\right], \tag{17}$$

where $|M|$ is the number of triplets and $|L|$ is the number of labels. For each triplet m at round d, the fully connected layer outputs $\mathbf{s}_{m,d} \in \mathbb{R}^{|L|}$, where $s_{m,d,k}$ denotes the logit score for label k. The predicted probability is obtained by applying the sigmoid function element-wise, i.e., $\hat{y}_{m,d,k} = \sigma(s_{m,d,k})$ with $\sigma(x) = \frac{1}{1+e^{-x}}$, and $y_{m,d,k} \in \{0, 1\}$ is the corresponding ground-truth indicator. At each round d, our ultimate optimization goal is to maximize the ELBO $\mathcal{L}_d$ in Eq. 11 of the topic learning module while minimizing the $loss_d$ in Eq. 17 of label prediction module:

$$loss_{total} = -\mathcal{L}_d + \gamma loss_d, \tag{18}$$

where γ is a hyperparameter.

4. Experiments

In this section, we conduct several experiments with real-world datasets to validate our ideas and evaluate our proposed framework.

4.1. Data preprocessing

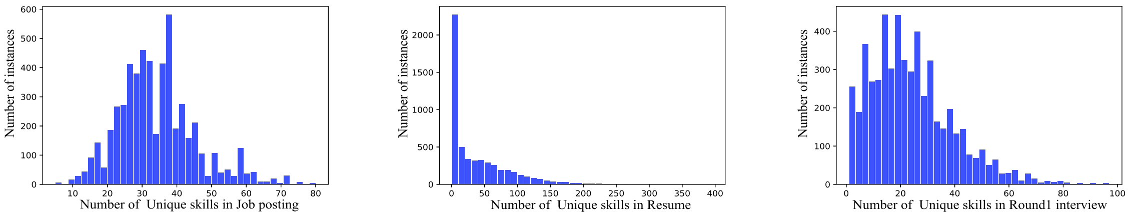
4.1.1. Data description

Our experiments use exemplary interview cases collected by the HR department of a leading high-tech company. The company follows standardized and rigorous hiring practices, making this data source representative. Candidates recruited based on these cases have shown better post-entry performance and meet the “person-job fit” standard. Accordingly, the assessment points recorded in these interviews are reliable for evaluating candidates. Learning from these patterns, our model proposes high-quality skill assessment points. The source data include three datasets: a position dataset, an applicant dataset, and an interview report dataset. Table 1 reports the volume and information of each dataset. The position dataset contains job advertisements. The applicant dataset contains resumes. The interview report dataset contains interviewer reports.

To focus on technical positions, we removed non-technical positions from the position dataset. Next, we linked the three datasets by application ID to form triplets. Each application ID corresponds to one job posting ID and one resume ID. It may be associated with multiple interview IDs because interviews are conducted in multiple rounds. We merged interview reports that share the same application ID. We then retained only triplets with complete records for all three rounds. The final dataset includes 5437 triplets, covering 1573 job postings and 4874 resumes. In our experiments, we focus on textual information in all datasets. We treat job description and job content as the job posting profile, project experience as the resume profile, and evaluation records as the interview profiles. Each sample is formatted as *Application ID, Job posting, Resume, Round1 interview record, Round2 interview record, Round3 interview record.*

Table 2
The statistics of the skill words.

Data type	NO. of skill vocabulary	Length of skill sets per instance		
		Repetitive	Non-repetitive	Unique
Job posting	2,796	39.14	34.19	-
Resume	6,320	63.67	38.80	-
Round1 interview	5,146	38.58	27.03	23.9
Round2 interview	5,146	34.11	24.59	21.85
Round3 interview	5,146	26.59	20.03	17.91



(a) (b) (c)

Fig. 4. The distribution of the size of skill sets in Job posting (a), Resume (b) and Round1 interview record (c).

(a) (b) (c)

Fig. 5. The skill word cloud of Job postings (a), Resumes (b) and the total three rounds of Interview records (c).

4.1.2. Skill word extraction

As introduced in Section 3.1, this work focuses on skill inspection in intelligent interview assistance. Therefore, our first step is to identify skill-related content. By definition, skill keywords are phrases frequently used to evaluate candidates in interviews. They include not only common skills but also key assessment points that interviewers care about. Following prior work Qin et al. (2019), we train an LSTM-CRF model to extract skill phrases from the original text, including *Job posting, Resume, Round1 interview record, Round2 interview record, and Round3 interview record*. To construct ground-truth skill words for training and validation, we manually annotate a subset of instances using an online annotation platform. The LSTM-CRF achieves an F1-score of 0.72 on application text and 0.62 on interview text. Next, we invite 10 human-resources experts to filter the extracted phrases based on our definition and their professional judgment. Using the filtered phrases, we build separate skill vocabularies for job postings, resumes, and interview records. We then filter each raw text with its corresponding vocabulary. We keep the original order of the retained skill words, so the resulting skill sets preserve the natural sequential context.

Table 2 reports descriptive statistics of the skill vocabulary and skill sets for each text type. The interview skill vocabulary contains 5146 skills, which serve as prediction labels in subsequent experiments. The repetitive set keeps repeated skills within an instance, considering that their different sequence positions involve different semantic relationships. The non-repetitive set removes duplicates within an instance. The unique set retains only skills that appear in interview records but not in job postings or resumes. As shown by the size of the unique set in Table 2, many skills are unique to interview rounds, which supports the expression-gap issue discussed earlier.

We depict the distribution of skill set length in Fig. 4 and the skill word distribution in Fig. 5. The word cloud further illustrates our first challenge: the expression gap. Overall, there is not much difference in the expression of skills between job posts and resumes. However, it is far more different in the interview, which pays more attention to the detailed knowledge of certain skills, such as “function”, “array”, and “storage”. These terms are rarely stated in job postings or resumes, but they are frequently examined during interviews. They also reflect basic knowledge in programming languages and data structures. Together, these results suggest that job postings, resumes, and interviews share related semantic topics, but differ in how skills are expressed.

In addition, another innovation of this paper is the multi-interview suggestion. To confirm the necessity of multiple rounds, we counted the word frequencies of each skill word in the three rounds of interviews respectively, denoted as $F_i (i \in \{1, 2, 3\})$. Then each skill word’s probability of occurrence in each round could be calculated as $P_i = F_i / \sum_{i=1}^{K=3} F_i$. We screened out the top 20 words with

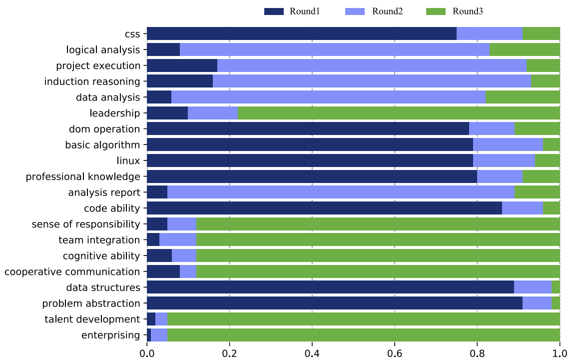


Fig. 6. Skill words in multi-round interviews.

the largest variance of three probability values and provided the Probability Distribution graph. As shown in Fig. 6, in the Round1 interview, there are many basic technical words such as "css", "dom operation", "linux", and "data structure". In Round2 interview, more emphasis is placed on reasoning, analysis, and execution, such as "logical analysis", "data analysis", "induction reasoning", etc. The Round3 interview focuses more on soft power, such as "leadership", "sense of responsibility", and "enterprising", which pay more attention to talent development.

4.2. Experimental setup

4.2.1. Comparison methods

To evaluate our model, we compare it with representative baselines. Our work makes two technical contributions: (1) combining label graphs with NTMs for representation learning, and (2) introducing a topic-attention mechanism that leverages topic parameters inferred by a GSM-based NTM for multi-label tasks. We include baselines from different modeling families, including LSTM (Liu et al., 2016), LSTM-AT (Zhou et al., 2016) (which incorporates an attention mechanism), and DocBERT (Adhikari, 2019) (a BERT-based method). Additionally, we included methods specifically designed for extreme multi-label classification, such as XMLCNN (Liu et al., 2017) and BertXML (Xun et al., 2020). We also considered methods that incorporate label information, such as LAHA (Huang et al., 2022), a method that integrates word and label embedding via a hybrid-attention mechanism, and LR-GCN (Vu et al., 2023), which introduces a correlation matrix to model label correlations based on occurrence and co-occurrence probabilities. LR-GCN learns label representations via GCNs and document representations via RoBERTa. Considering the recent advances in large language models (LLMs), we further included two LLM foundation model baselines: DeepSeek-V3 (Liu et al., 2024), a proprietary LLM optimized for complex reasoning and multi-tasking; and Qwen-Plus, an open-source series from Alibaba Cloud known for its excellence in long-context understanding and structured data prediction. Unlike supervised models, these LLMs require no fine-tuning and rely solely on prompt-based inference, which reveals the comparison of generative approaches and task-specific representation learning. For both models, we adopt a one-shot prompting setup. The input contains only the application information (j and r). Under this setup, the model is provided with one fixed example sampled from the training split and is instructed to output a set of interviewer assessment skills, i.e., skill keywords that should be assessed during the interview. The output is restricted to keywords only (no explanations), and we use temperature sampling with $temperature = 0.7$. Our TOSM uses GSM (Miao et al., 2017) as the prototype. To isolate the effects of our components, we design two ablations: GSM-G, which uses the label graph and NTM representations for prediction without topic parameters; and GSM-AT, which removes the label graph and predicts labels using GSM-learned parameters only. Except for DocBERT and LR-GCN, which use BERT pre-trained models, all supervised baselines share 300-dimensional GloVe embeddings (Pennington et al., 2014) pre-trained on our corpus. We use binary cross-entropy loss for multi-label classification.

4.2.2. Evaluation metrics and experimental settings

We split our experiment dataset in the ratio of 6:2:2 into three subsets: the training set, the validation set and the test set. The validation set is used for parameter selection and the test set is used for performance evaluation.

In practice, we constructed skill graphs of interviews with window size $l = 3$. We set the Dirichlet prior parameter $\alpha = 1$. Batch-normalization (Ioffe & Szegedy, 2015) has been used on the inference network for the posterior parameters of θ. We utilized 300-dimensional GloVe word embeddings (Pennington et al., 2014) to fix X (i.e., $L = 300$) and word vectors in the graph encoder. In the LSTM, we set the dimension of the hidden state as 128 and the number of layers as 2. For the optimization, the Adam optimizer was used with the initial learning rate of 0.001 and the linear learning rate decay trick to find the optimal. Our code is available at `https://github.com/suyuhanzz/TOSM/tree/main`.

As we regard the skill assessment suggestion problem as a multi-label learning issue, we select the precision at top k ($P@k$), and the normalized discounted cumulative gain ($NDCG@k$) to evaluate the performance of these methods. The metric of $P@k$ is capable of measuring intuitively if the top-k relevant labels are presented at the real label set. The metric of $NDCG@k$ illustrates the quality of label ranking. For these metrics, the higher the value, the better the performance.

4.3. Experimental results

4.3.1. Predictive performance comparisons

We conducted five repeated experiments with different random seeds for all supervised models, and report the mean ± standard deviation across $P@k$ and $NDCG@k$ for $k = 5, 15, 30$. For LLM baselines, decoding is stochastic under temperature sampling. We therefore run each LLM baseline five independent times with identical prompts and hyperparameters, and report the mean ± standard deviation. In addition, we performed two-tailed t-tests to compare each baseline method with TOSM for every metric.

Table 3 and Table 4 report $P@k$ and $NDCG@k$ on the three-round interview dataset. We use *, **, and *** to denote $p < 0.1$, $p < 0.05$, and $p < 0.01$, and the best result in each column is highlighted in bold. Overall, TOSM significantly outperforms most baselines. The gains are most pronounced at $k = 30$ (i.e., $P@30$ and $NDCG@30$), especially in Rounds 2 and 3, suggesting that TOSM better retrieves a broader set of relevant skills (each label set contains about 20 skill-related words) and handles the increased complexity of later-round prediction. Among strong supervised baselines, DocBERT substantially improves over basic RNN and CNN methods (LSTM, LSTM-AT, and XMLCNN) but still falls short of TOSM, despite TOSM using only GloVe embeddings. LR-GCN is the most competitive baseline: it achieves superior results at $k = 5$ but is weaker at $k = 30$, indicating that it is more effective at predicting only the top few labels, whereas TOSM better recovers a wider range of relevant skills. The LLM baselines (DeepSeek-V3 and Qwen-Plus) perform consistently worse across rounds and metrics, highlighting the limitation of prompt-only inference without fine-tuning for this domain-specific extreme multi-label task. LAHA also performs poorly, suggesting that traditional embedding-based and label-aware attention approaches are insufficient for interview-point prediction. The ablation results further validate our design. GSM-G outperforms LAHA, confirming the value of the skill graph. In contrast, GSM-AT shows diminishing significance as rounds progress (highly significant in Round1 but less so in Round3), underscoring the superiority of the proposed topic-attention mechanism for multi-round prediction.

4.3.2. Model analysis

To further demonstrate the significance of skill graph incorporation and the topic attention mechanism to the model, we visualize some stage results from the perspective of graph, topic, and attention mechanism.

- **Skill Graph View.** Regarding the fact that there will be some dependencies between skill words in the preceding and following positions in a sentence, we build a skill relationship graph to capture the semantically related skill words from the interview record. To demonstrate the significance of this graph structure in topic learning, we visualized a subgraph of our construction. As shown in Fig. 7(a), "deep learning" is associated with "image", and skill words related to image processing such as "filter" and "feature selection" also establish edge relationships with them. "Project development" is not associated with "network equipment" and "data mining" on the surface semantics, but their frequent appearance in the interview record indicates that they may have some potential relationships for the interview skill evaluation. Thus, skill graphs attach skill words that are not semantically connected on the surface level via co-occurrence relationships, making it more informative for topic learning than common BOW vectors. For interviewers, the skill graph offers a view of skill relatedness: highly connected nodes support deeper follow-up questioning, while weakly connected nodes help diversify questions and broaden coverage.
- **Topic Vector View.** To verify the distinction between the three rounds of interview assessments and justify the need for multi-round interview prediction, we visualized the trained topic vectors from the three rounds using t-SNE to convert the original 300-dimensional vectors into a 2-dimensional space (see Fig. 7(b)). Different shapes represent topics from different rounds. The visualization shows that topics from the three rounds differ, with second-round topics being relatively concentrated, indicating similar semantics. In contrast, third-round topics are the most dispersed, indicating more diverse semantics. The clear separation between the third and the other rounds validates our approach to multi-round interview suggestions. Additionally, comparison results show that metrics decline round by round due to increased task difficulty from greater topic diversity.
- **Topic Distribution View.** We visualize the topic distribution, $\theta_{d,k}$, to demonstrate how topics are tailored for specific applications. We randomly selected ten application instances from the dataset, visualizing their topic distributions in Fig. 8, where darker colors indicate higher values. The results show that most samples have the highest probability for Topic 4 in the first round and Topic 10 in the second round. We then selected three topics per round and listed their top words by probability. In practice, displaying the dominant topics and their representative skill words can guide interviewers in selecting round-specific assessment points by clarifying which skill themes should be prioritized in each round. The m-th topic of the n-th round is abbreviated as Topic $n - m$. As shown in Table 5, Topic 1–4 and Topic 2–19 mainly cover machine learning and deep learning algorithms, while Topic 1–19 and Topic 2–10 focus on database operations and maintenance. Fig. 8 reveals that in the first round, Topic 1–4 for Sample 6 and Topic 1–19 for Sample 8 are prominent; in the second round, Topic 2–4 and Topic 2–19 for Sample 6 and Topic 2–10 for Sample 8 are significant. Table 6 shows that Sample 6, a data mining engineer, has skills like "algorithm design", "neural network", and "decision tree", aligning with Topics 1–4, 2–4, and 2–19. Sample 8, a system development engineer, has "mysql" and "storage" skills, matching Topics 1–19 and 2–10. This confirms the effectiveness of $\theta_{d,k}$ in topic learning, ensuring personalized topics for specific applications.
- **Topic Attention View.** With the topic attention mechanism in TOSM, we can easily capture the different topic semantic focus of the same application in different interview rounds. Here we randomly select 30 samples from the dataset and visualize the accumulated sum topic attention $\alpha_{d,t}$ learned by the topic attention mechanism for each skill word. As shown in Fig. 9. Detailed technical knowledge like "message queue" was given strong attention in Round2 while "teamwork ability", and "learning ability" got more attention in Round3. The generalized descriptions like "machine learning" were given importance in Round1 while

Table 3
Performance results in *P@k* for 3 rounds skill suggestion.

	P@5	p-value	P@15	p-value	P@30	p-value
Round1:						
LSTM	0.1578±0.0003	***	0.1360±0.0006	***	0.1195±0.0005	***
LSTM-AT	0.1557±0.0032	***	0.1360±0.0013	***	0.1196±0.0004	***
XMLCNN	0.1552±0.0028	***	0.1351±0.0008	***	0.1179±0.0013	***
BertXML	0.2222±0.0042	***	0.1820±0.0021	***	0.1455±0.0018	***
DocBERT	0.2531±0.0054		0.1932±0.0021	**	0.1584±0.0017	**
LAHA	0.1440±0.0088	***	0.1310±0.0035	***	0.1165±0.0026	***
LRGCN	**0.2654±0.0077**	**	0.1943±0.0020		0.1597±0.0014	*
DeepSeek-V3	0.0952± 0.0014	***	0.0737±0.0023	***	0.0496±0.0013	***
Qwen-Plus	0.1184±0.0016	***	0.0867±0.0016	***	0.0609±0.0015	***
GSM-G	0.1999±0.0005	***	0.1627±0.0009	***	0.1391±0.0019	***
GSM-AT	0.2383±0.0078	***	0.1874±0.0041	***	0.1548±0.0024	***
TOSM	0.2540±0.0029		**0.1971±0.0027**		**0.1623±0.0023**	
Round2:						
LSTM	0.1271±0.0019	***	0.1033±0.0006	***	0.0913±0.0004	***
LSTM-AT	0.1280±0.0023	***	0.1032±0.0008	***	0.0916±0.0005	***
XMLCNN	0.1259±0.0025	***	0.1016±0.0013	***	0.0881±0.0014	***
BertXML	0.1801±0.0028	***	0.1436±0.0024	***	0.1119±0.0022	***
DocBERT	0.1924±0.0017	***	0.1474±0.0022	***	0.1218±0.0013	***
LAHA	0.1280±0.0024	***	0.1028±0.0006	***	0.0913±0.0005	***
LRGCN	**0.2068±0.0022**		0.1490±0.0013	***	0.1225±0.0013	***
DeepSeek-V3	0.0788±0.0014	***	0.0595±0.0010	***	0.0436±0.0006	***
Qwen-Plus	0.0997±0.0004	***	0.0700±0.0007	***	0.0500±0.0007	***
GSM-G	0.1548±0.0026	***	0.1270±0.0026	***	0.1089±0.0025	***
GSM-AT	0.1929±0.0043	*	0.1476±0.0020	***	0.1206±0.0015	***
TOSM	0.1972±0.0027		**0.1547±0.0021**		**0.1272±0.0011**	
Round3:						
LSTM	0.1711±0.0000	***	0.1228±0.0010	***	0.0971±0.0004	***
LSTM-AT	0.1711±0.0000	***	0.1223±0.0005	***	0.0976±0.0003	***
XMLCNN	0.1668±0.0059	***	0.1227±0.0013	***	0.0949±0.0012	***
BertXML	0.1830±0.0017	***	0.1353±0.0015	***	0.1051±0.0010	***
DocBERT	0.1857±0.0026	*	0.1387±0.0023	***	0.1067±0.0013	***
LAHA	0.1696±0.0033	***	0.1228±0.0010	***	0.0969±0.0019	***
LRGCN	0.1877±0.0025	***	0.1420±0.0010	**	0.1089±0.0007	***
DeepSeek-V3	0.0286±0.0007	***	0.0195±0.0004	***	0.0131±0.0004	***
Qwen-Plus	0.0508±0.0018	***	0.0386±0.0007	***	0.0303±0.0006	***
GSM-G	0.1795±0.0017	***	0.1317±0.0007	***	0.1053±0.0008	***
GSM-AT	0.1874±0.0023		0.1408±0.0009	**	0.1084±0.0009	***
TOSM	**0.1896±0.0037**		**0.1458±0.0034**		**0.1117±0.0016**	

"machine learning algorithms" such specific descriptions were emphasized in Round2. This completely demonstrates that in a topic-oriented multi-label learning method, different rounds of attention mechanisms focus on different skill words, achieving progressive suggestions for multiple rounds. Such attention patterns make the model's round-specific focus interpretable, helping interviewers understand why skill keywords are recommended and plan targeted questions accordingly.

4.3.3. Hyperparameter analysis and robustness check

Finally, we test the tuning of two important hyperparameters in our experiments: the number of topics in the topic attention mechanism and the window size for establishing the skill edge relationship in the construction of the skill graph. As shown in Fig. 10(a), when the number of topics is between 0 and 50, the model's performance is stable. However, as the number of topics increases, the model effect may decrease. Based on the results, it can be inferred that the number of topics in the interview-related text data is less than 50. Fig. 10(b) showed that the best performance was achieved when window size = 3, suggesting that potential relationships between moderately neighboring skill words are more effective for prediction. Additionally, when the number of topics is below 50, we compare our model with other topic model comparison methods. As shown in Fig. 10(c), our approach outperforms both GSM-G and GSM-AT for each number of topics.

4.3.4. Application case for interview question generation

To conduct the experiment, we utilized the same prompt, "Help me ask 3 interview questions for each of the three rounds of interviews based on the following skill keywords." We extracted key skill words from both the original application information and our suggested key skill words. Using these skill words, we generated interview questions with ChatGPT 3.5.

The generated questions were then compiled into a table for comparison. Next, we conducted a field experiment involving 10 interviewers. We provided both versions of the questions to the interviewers simultaneously and asked them to choose the version

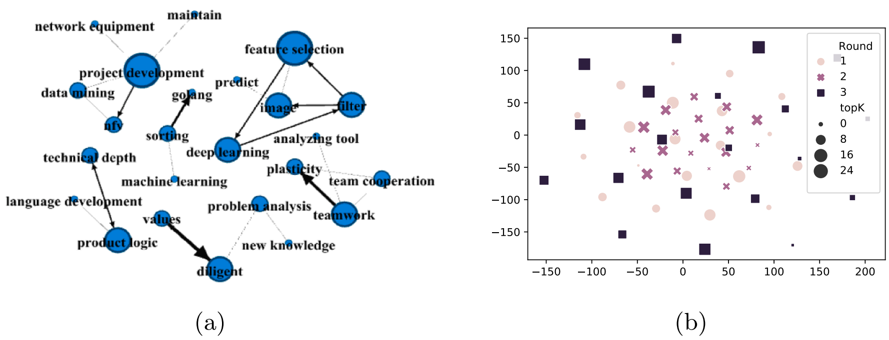


Fig. 7. Visualization of (a) the skill graph structure and (b) t-SNE projection of round-specific topic vectors. Different markers indicate topics learned from different interview rounds, illustrating round-level semantic separation.

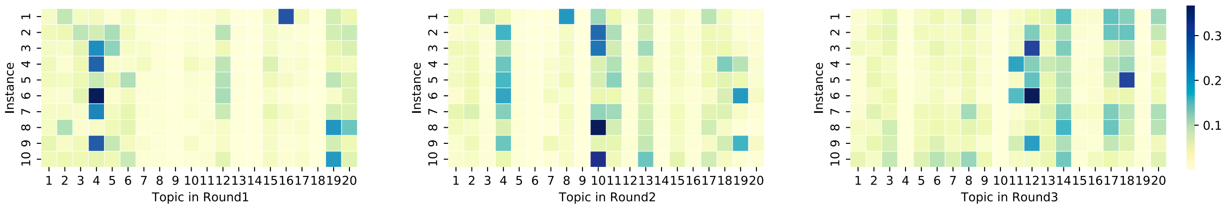


Fig. 8. Topic proportion heatmaps for 10 sampled applications. Darker cells indicate higher inferred topic probability, showing application-specific topic profiles.

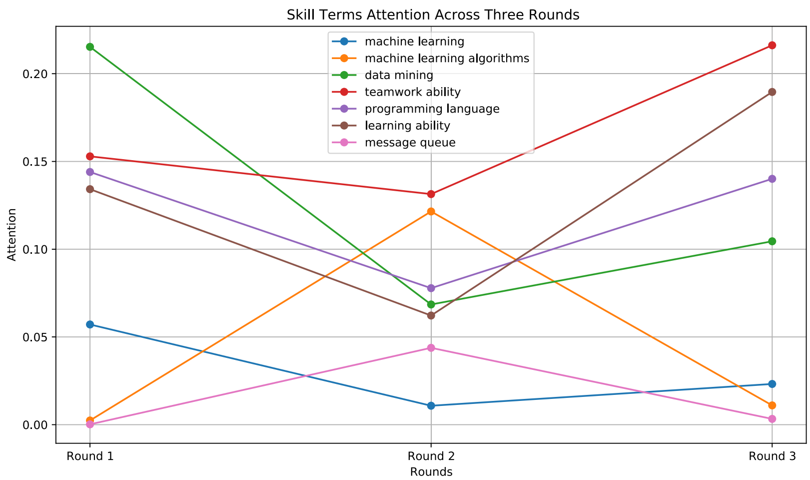


Fig. 9. Topic-attention aggregation over skill keywords across rounds. The summed attention weights $\alpha_{d,t}$ highlight which skills are most emphasized in each round (30 samples).

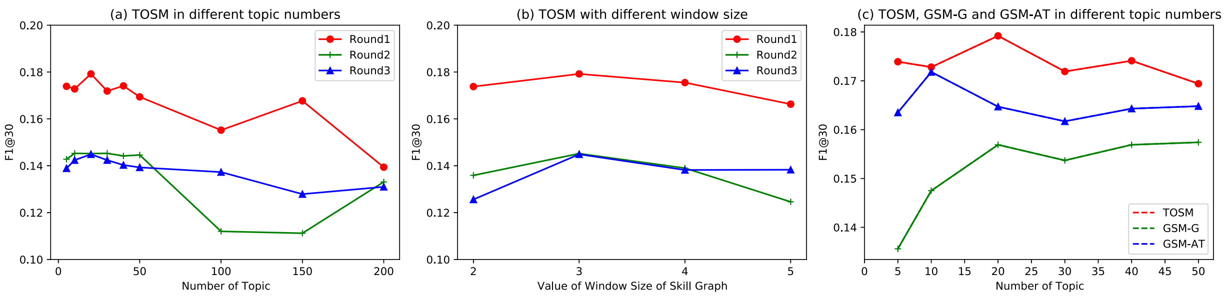


Fig. 10. F1@30 performance with hyperparameter analysis and robustness check.

Table 4
Performance results in *NDCG@k* for 3 rounds skill suggestion.

	NDCG@5	p-value	NDCG@15	p-value	NDCG@30	p-value
Round1:						
LSTM	0.1568±0.0006	***	0.1476±0.0007	***	0.1529±0.0005	***
LSTM-AT	0.1554±0.0020	***	0.1475±0.0018	***	0.1530±0.0006	***
XMLCNN	0.1551±0.0029	***	0.1456±0.0011	***	0.1513±0.0012	***
BertXML	0.2293±0.0032	***	0.2013±0.0022	***	0.2002±0.0018	***
DocBERT	0.2657±0.0025		0.2199±0.0034	***	0.2208±0.0031	
LAHA	0.1492±0.0057	***	0.1402±0.0043	***	0.1468±0.0042	***
LRGCN	**0.2752±0.0029**	*	0.2290±0.0032		0.2210±0.0025	
DeepSeek-V3	0.0976±0.0014	***	0.0855±0.0018	***	0.0755±0.0012	***
Qwen-Plus	0.1210±0.0009	***	0.1017±0.0014	***	0.0914±0.0016	***
GSM-G	0.2102±0.0068	***	0.1840±0.0028	***	0.1821±0.0030	***
GSM-AT	0.2537±0.0079	***	0.2170±0.0051	***	0.2119±0.0040	***
TOSM	0.2681±0.0036		**0.2297±0.0021**		**0.2233±0.0025**	
Round2:						
LSTM	0.1308±0.0016	***	0.1174±0.0005	***	0.1260±0.0008	***
LSTM-AT	0.1316±0.0015	***	0.1175±0.0009	***	0.1263±0.0001	***
XMLCNN	0.1288±0.0038	***	0.1155±0.0012	***	0.1224±0.0020	***
BertXML	0.1765±0.0035	***	0.1606±0.0021	***	0.1623±0.0021	***
DocBERT	0.2013±0.0048	**	0.1707±0.0029	***	0.1761±0.0023	***
LAHA	0.1313±0.0019	***	0.1182±0.0007	***	0.1258±0.0008	***
LRGCN	**0.2142±0.0016**	***	0.1784±0.0013		0.1809±0.0006	**
DeepSeek-V3	0.0845±0.0015	***	0.0729±0.0005	***	0.0703±0.0005	***
Qwen-Plus	0.1070±0.0014	***	0.0877±0.0006	***	0.0826±0.0004	***
GSM-G	0.1602±0.0027	***	0.1439±0.0037	***	0.1506±0.0031	***
GSM-AT	0.2056±0.0047		0.1736±0.0022	***	0.1753±0.0022	***
TOSM	0.2093±0.0023		**0.1797±0.0013**		**0.1829±0.0008**	
Round3:						
LSTM	0.1745±0.0018	***	0.1568±0.0009	***	0.1663±0.0008	***
LSTM-AT	0.1738±0.0017	***	0.1565±0.0007	***	0.1666±0.0003	***
XMLCNN	0.1712±0.0054	***	0.1560±0.0007	***	0.1629±0.0009	***
BertXML	0.1901±0.0035	***	0.1701±0.0025	***	0.1810±0.0015	***
DocBERT	0.1961±0.0023		0.1766±0.0016	***	0.1834±0.0017	***
LAHA	0.1258±0.0008	***	0.1625±0.0016	***	0.1578±0.0020	***
LRGCN	**0.2038±0.0032**		0.1823±0.0031		0.1894±0.0019	***
DeepSeek-V3	0.0336±0.0013	***	0.0272±0.0008	***	0.0253±0.0008	***
Qwen-Plus	0.0602±0.0021	***	0.0526±0.0012	***	0.0534±0.0013	***
GSM-G	0.1871±0.0012	***	0.1667±0.0004	***	0.1765±0.0004	***
GSM-AT	0.1991±0.0023		0.1790±0.0014	*	0.1860±0.0017	***
TOSM	0.2005±0.0052		**0.1828±0.0036**		**0.1938±0.0016**	

they preferred. The questions details are in Fig. 11. Surprisingly, 90% of the interviewers selected the version based on our suggested words. This finding indicates that the questions aligned more closely with the desired interview style. This outcome provides empirical evidence supporting the effectiveness of our TOSM in facilitating interview questioning. The higher preference for questions generated with our suggested key skill words underscores their relevance and appropriateness for conducting interviews.

5. Discussions

This study proposes a novel TOSM for AI-driven multi-round interview questioning. The framework comprises two core components: the Graph-based Topic Learning Module and the Topic-specific Keyword Suggestion Module. Using a real exemplary interview case dataset from a leading high-tech company, we train the model and evaluate TOSM's performance on skill keyword suggestion tasks. Compared with various baselines, TOSM achieves significantly higher *precision* and *NDCG* scores when predicting a broader range of relevant labels and in later rounds, demonstrating overall superiority in multi-round interview settings. Visualization analyses further confirm the interpretability of TOSM. The skill graph shows how graph-based topic learning bridges vocabulary mismatches between skill expressions in application information and interview questioning. Topic distribution and attention visualizations demonstrate that the model adapts its focus to each interview round through the topic attention mechanism. Robustness checks on hyperparameters confirm the model's stability and generalizability. Finally, we assess TOSM's practical utility by using its suggested skill keywords for question generation via ChatGPT. Feedback from professional interviewers indicates that these questions are more accepted than those generated by directly extracting words from application materials. Overall, TOSM provides interviewers with context-aware skill assessment guidance that can be refined across multiple interview rounds.

Table 5
Topics of skills.

Round1	
Topic4.	c++, programming, machine learning, recognition, automation, deep learning, library, c language, lstm, compilation, cnn
Topic10.	hardware, driver, switch, signal, digital signal processing, wifi, ospf routing protocol, data flow, bandwidth, network card
Topic19.	database, interface, multithreading, mysql, process, php, operation and maintenance, redis, storage, spring,cache, log, monitoring, concurrency
Round2	
Topic4.	machine learning, deep learning, big data programming ability, logical thinking, understanding ability, data analysis, data mining, algorithmic ability
Topic10.	database, index, storage, mysql, redis, monitoring, operation and maintenance, sql, log, spring, cache, concurrency
Topic19.	gpu, image, cnn, speech recognition, extraction, neural network, lda, gbdt, svm, nlp, convolution, clustering, machine learning, sampling
Round3	
Topic4.	system, computer software, management system, struts, software development engineer, storm, cluster, open source framework
Topic10.	communication skills, learning skills, teamwork, logical thinking skills, ai, new technologies
Topic19.	vr, sdn, gps, vpn, system development, mobile client

Table 6
The keywords in application information of Sample6 and Sample8.

Sample6–Data Mining Senior Engineer	Sample8–System Development Engineer
c, c++, python, linux platform, shell programming, data structure, algorithm design, data mining, algorithm, neural network, decision tree, svm, clustering, logistic regression, bayesian, recommendation algorithm, deep learning, nlp, communication skills, teamwork spirit	system development, linux, programming, struts development framework, web development technology, script programming, shell, python, multi-threaded programming, open source, storage system, mysql

	ChatGPT3.5 generated questions based on application information keyword	**ChatGPT3.5 generated questions based on suggestion keyword**
Round1	1. Please describe an example of a difficult problem encountered in a computer vision or natural language processing project and detail how you went about the process of problem analysis and solution. 2. For image processing and image classification tasks, have you ever used Convolutional Neural Networks (CNN)? Please explain how CNNs work in image classification and talk about the challenges you encountered in practical applications and how you overcame them. 3. Deep learning has a wide range of applications in computer vision and natural language processing. Can you share some of your knowledge about deep learning algorithms or models and give examples of how to choose the right deep learning model to solve a problem in a project?	1. Please tell us about your experience in Kaggle data science competitions and the projects you have participated in. How did you apply machine learning algorithms and data science techniques to solve real-world problems? 2. For tasks like similar articles recommended, you can introduce the commonly used recommendation algorithms and principles, and discuss their advantages and disadvantages in practical applications. 3. For Natural Language Processing (NLP), can you explain in detail the principles of Word2Vec and CRF algorithms and share some of your experiences in applying them in NLP projects?
Round2	1. In image processing, feature extraction is crucial for tasks such as image classification. How do you extract image features when processing images? Please talk about the methods you use and their advantages and disadvantages. 2. Word2Vec in natural language processing is a commonly used method for representing word vectors. Please explain the principle of Word2Vec and how it can be applied in your project to improve the performance of NLP tasks. 3. Do you have experience competing in Kaggle data science competitions? If so, please share a competition project in which you participated and the technique, algorithm, or model used in the competition. If not experienced in competitions, how do you evaluate the merits of an algorithm or model and make decisions in your project?	1. In terms of programming, do you have experience in solving data structure and algorithmic problems? Please write an algorithm for reverse array (reverse array) and analyze its time complexity. 2. Are you proficient in programming in a scripting language? Write a program in a scripting language that counts the number of occurrences of each word in a given text and sorts the results in order of frequency. 3. In machine learning, you have mentioned Word2Vec, Random Forest, GBDT etc. Can you explain the fundamentals of these algorithms and talk about how to choose appropriate evaluation methods and common loss functions in different scenarios?
Round3	1. Code programming and code quality in a project are critical to achieving efficient algorithms and models. Please talk about how you ensure code readability, maintainability, and high performance in your projects. 2. In deep learning and machine learning projects, the ability to understand and make decisions has a big impact on success. Please share a project where you had to make important decisions and how you made them based on data and analytics. 3. In the future, do you plan to continue to work deeply in computer vision, natural language processing, or other areas? Please talk about your career plans and learning plans, and how you can synthesize multiple skills from the skill keywords for your future projects.	1. Please talk about your understanding of the basic concepts and application scenarios of algorithms such as HMM, CRF and LSTM. Have you applied these algorithms in real projects? If so, please share your experience and results. 2. In terms of machine learning basics, what do you think are the most important? Please talk about your learning process and how you maintain your ability to learn and keep up with new research and algorithms. 3. In terms of engineering skills and problem solving, can you share some of the technical challenges you have faced in your projects and how you solved them? Please describe a project experience that you feel is most representative of your experience and the intelligence and problem solving skills you demonstrated on that project.

Fig. 11. Comparison of interview question generation based on ChatGPT 3.5.

5.1. Theoretical contributions

This study makes several theoretical contributions. First, it contributes to the HRM and employment interview literature by introducing an AI-based approach for skill assessment in multi-round interviews. Prior work suggests that unstructured interviews are generally less reliable and valid than structured interviews (Judge et al., 2000). Building on this insight, we formulate multi-round interviewing as a setting that requires round-specific, application-relevant assessment targets, and propose an AI framework that provides application-aware skill keyword suggestions to support structured skill coverage across rounds.

Second, this study extends AI-driven recruitment research by shifting attention from pre-interview recommendation and candidate-side assessment to interviewer-side decision support during the interview. Most prior work targets resume screening, person-job matching, and job recommendation, or predicts interview scores and provides feedback for candidates. In contrast, interviewers face a different decision problem: determining what skills and questions to assess under time constraints and multi-round progression. By formulating interviewer-side skill keyword suggestion as the core task, the TOSM framework addresses two key challenges in AI-assisted interviewing: bridging semantic gaps between application information and interview content, and adapting to the multi-round progression of interviews. Furthermore, this research contributes to the growing literature on human-AI collaboration (Li et al., 2025). While previous studies have examined human-AI interaction in investment and medical diagnosis (Ge et al., 2021; Schick & Fischer, 2021), the scenario in interview remain limited. Our framework highlights the potential of extending human-AI collaboration theory to interviewer question design and opens new directions for examining the effectiveness of AI assistance in HRM area (Park et al., 2021; Zhai et al., 2024).

5.2. System design and practical implications

This study proposes an AI framework that integrates graph-based topic learning with topic-oriented multi-label learning for multi-round interview assessment suggestion. Methodologically, it augments NTMs with graph-based representation learning to capture latent associations among heterogeneous recruitment texts, and introduces a topic attention mechanism to support round-specific keyword suggestion as interview focus evolves across rounds.

From a practical perspective, TOSM can serve as a building block for AI-assisted interview systems that improve the efficiency and structure of skill assessment. First, it can support question recommendation by retrieving or generating suitable questions from a question bank based on job requirements, candidate profiles, enabling more structured, targeted, and effective assessments, reducing subjectivity. Second, it can support interviewer assignment by leveraging historical interview records to infer interviewer expertise and questioning patterns, thereby facilitating a more appropriate interviewer-candidate matching system for multi-round evaluations.

Beyond recruitment applications, the methodological framework developed in this study can be extended to other recommendation domains that involve heterogeneous but semantically related textual resources. For example, in the medical domain, diagnostic decision-making and treatment planning could employ a comparable cross-text learning mechanism to capture semantic relationships between pathology reports and clinical guidelines. This extensibility highlights the framework's potential for addressing complex decision-support tasks that require semantic integration across diverse information sources.

5.3. Limitations and future work

Our study also has some limitations that can be explored in future research. Based on the experimental results, we find that TOSM and the existing models do not perform well for the task of skill assessment keyword suggestion, even though TOSM is improved. It shows that the algorithmic exploration in this field is still relatively elementary, therefore, there are the following points that can be improved from the methodological point of view. First, for skill extraction, we used a keyword extraction model based on LSTM-CRF to construct a skill vocabulary list and then used vocabulary matching to filter the text. However, in this way, some discontinuous skill expressions cannot be extracted, such as "java programming", which is represented as "programming with java" in the text. We can explore a better method to extract skill labels. Second, in terms of skill keyword suggestion, we only consider the co-occurrence relationship of skills in this paper. We can further explore the deep relationship between skills, such as the inclusion relationship and recurrence relationship, to achieve a cascading effect when suggesting, for example, asking questions about parent knowledge points first and then asking questions about sub knowledge points. Third, in prior-posterior co-adjustment of the joint topic semantic learning, we directly merge the job posting and resume as application information to be the prior and the historical interviews as the posterior. No distinction is made between the job information and the resume text. However, there is also a certain expression gap between them. So further processing of semantic association mining between them is needed in the future. Fourth, our current formulation does not model interactive dynamics, since interviewer questioning in practice depends on candidates' responses; incorporating answer-aware interaction modeling would be valuable for interview-assistance agents. Fifth, in this paper, we mainly focus on technical positions as it is clearer to identify skill terms and the skills relationship by word extraction and graph building. However, for non-technical positions, there is a certain degree of difficulty as skill extraction is more difficult to realize. More experts need to participate to help define the keyword for non-technical. It is also of great significance to further research on interview questioning patterns of different position types.

Besides methodological limitations, data generalizability is another limitation. Our dataset is sourced from a single technology company, and the learned interview patterns may encode organization-specific norms. Future work could validate the proposed framework on multi-organization or cross-industry datasets to assess robustness across different interview styles and evaluation standards. Finally, field studies could empirically examine the practical impact of AI-assisted questioning on downstream hiring outcomes (e.g., person-job fit and long-term performance), as well as adoption-related factors (e.g., when and why interviewers accept, edit, or reject AI suggestions). Fairness-related outcomes, such as candidates' perceived fairness and the balance of assessment points, also warrant explicit operationalization and empirical evaluation in future work.

CRediT authorship contribution statement

Yuhan Su: Methodology, Formal analysis, Visualization, Writing – original draft, Writing – review & editing; **Hongke Zhao:** Supervision, Project administration, Writing – review & editing; **Chuan Qin:** Conceptualization, Validation, Writing – review & editing; **Dazhong Shen:** Data curation, Methodology, Writing – original draft; **Hengshu Zhu:** Resources, Funding acquisition.

Data availability

The authors do not have permission to share data.

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